



KCA UNIVERSITY

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Department of Computing

PROJECT PROPOSAL

Predicting Developer Technology Decline Using Community Signals and Enterprise Adoption Patterns: A Machine Learning Approach

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Student Name: Tony Kenga

Registration Number: 24/03652

Supervisor: Dr. Rufus Gireka

ORCID: 0009-0007-6899-8590

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Faculty: School of Technology (SoT)

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ABSTRACT

Technology investment decisions, whether by organisations selecting enterprise software stacks or developers choosing skills to learn, carry substantial financial and career risk. When a technology enters a phase of silent decline, organisations face costly migrations and developers risk obsolescence, yet most stakeholders currently rely on trend articles and informal opinion rather than empirical evidence.

This proposal presents TechPulse, a supervised machine learning system that predicts the future trajectory of software technologies, classifying them as growing, stable, or declining, by analysing three distinct signal families: community activity derived from Stack Overflow question volumes and engagement rates; enterprise adoption depth extracted from Fortune 500 technology stack profiles; and developer sentiment drawn from annual survey data. TechPulse extends an existing, publicly starred open-source repository (developer-ecosystem-analytics) that already integrates six real-world datasets and twelve descriptive SQL analytics queries but currently offers no forward-looking predictive capability.

The project's objectives are to perform exploratory data analysis on the six-dataset corpus; to engineer predictive features capturing community decay, growth momentum, and adoption breadth; to train and evaluate four classification models (Logistic Regression, K-Nearest Neighbours, Random Forest, and XGBoost); to identify the strongest predictors of technology decline using SHAP feature-importance analysis; and to surface predictions and risk scores through an interactive Streamlit dashboard deployed publicly online.

The expected outcome is a functional, explainable prediction system that moves the developer-ecosystem-analytics repository beyond description into actionable forecasting, benefiting developers, organisations, and the broader open-source community immediately upon release.

CHAPTER ONE: INTRODUCTION

1.1 Background

Technology adoption decisions carry significant financial and strategic risk for both organisations and individual developers. Organisations that invest in declining technologies face costly platform migrations, retraining costs, and talent acquisition difficulties. Developers who specialise in technologies heading toward obsolescence face career displacement and diminished market value. In 2023 alone, organisations globally spent an estimated USD 4.7 trillion on information technology, and a significant proportion of that investment was allocated to software tools and platforms whose long-term viability was uncertain (Gartner, 2023).

At the global level, the pace at which technologies mature, stabilise, and eventually decline has accelerated. Technologies such as CoffeeScript, Apache Struts, and Perl experienced rapid community decline within five to ten years of peak adoption, often leaving organisations and developers with limited warning signals. Conversely, technologies such as Rust, TypeScript, and PostgreSQL have demonstrated sustained, measurable community growth, supported by strong enterprise adoption patterns and positive developer sentiment.

In the Kenyan and broader East African context, technology adoption decisions are particularly consequential. Many organisations, from fintech firms in Nairobi's Silicon Savannah to government digital transformation initiatives, invest in technology stacks with limited access to localised, evidence-based adoption guidance. Developers in the region similarly rely heavily on international trend publications, which may not reflect local enterprise adoption realities or the specific technologies dominating regional employer demand.

This project builds on an existing open-source repository, developer-ecosystem-analytics (19 GitHub stars), which already aggregates six real-world datasets covering Stack Overflow activity, developer sentiment surveys, Fortune 500 technology stacks, company profiles, and technology metadata. The repository currently implements twelve descriptive SQL analytics queries and a composite Technology Health Score, but does not yet offer any forward-looking, predictive capability. This gap, between descriptive community analytics and actionable technology forecasting, is the problem this project addresses.

1.2 Problem Statement

Organisations and individual developers currently lack a reliable, data-driven method for anticipating whether a given software technology is heading toward growth, stability, or decline. Existing community analytics tools, including the developer-ecosystem-analytics repository this project extends, are descriptive rather than predictive: they report what has already occurred in a technology's community and enterprise adoption activity, but they do not forecast where that technology is heading.

This limitation has measurable consequences. Organisations that commit engineering resources to technologies on a silent decline trajectory risk costly, unplanned migrations. Developers who invest significant learning time in a technology approaching obsolescence face career setbacks that may take years to recover from. Yet current decision-making remains largely dependent on informal opinion, trend articles, and developer surveys that describe present sentiment without projecting future trajectory.

The developer-ecosystem-analytics repository demonstrates that sufficient signal data exists, six integrated datasets spanning community activity, enterprise adoption, and developer sentiment, to support forward-looking analysis. However, no predictive layer currently exists to translate these signals into actionable trajectory classifications. This project addresses that gap directly by building TechPulse: a machine learning system that engineers features from existing signals and classifies technologies as growing, stable, or declining.

1.3 Proposed Solution

This project proposes developing TechPulse, a machine learning system that extends the existing developer-ecosystem-analytics repository with a predictive layer. TechPulse will:

- Engineer predictive features from community signals (Stack Overflow activity trends, question volumes, closure rates), enterprise adoption depth (Fortune 500 technology stack profiles), and developer sentiment data (satisfaction scores, adoption intent);
- Train and evaluate four classification models, Logistic Regression, K-Nearest Neighbours, Random Forest, and XGBoost, to label technologies as growing, stable, or declining;
- Apply SHAP (SHapley Additive exPlanations) analysis to identify and rank the strongest predictors of technology decline; and
- Present the resulting predictions and risk scores through an interactive Streamlit dashboard, deployed publicly via Streamlit Cloud.

TechPulse will extend rather than replace the existing repository, preserving all twelve descriptive SQL analytics queries and the composite Technology Health Score while adding a supervised prediction capability on top of them.

1.4 Objectives

1.4.1 Main Objective

To develop TechPulse, a machine learning system that predicts the future trajectory (growth, stability, or decline) of software technologies using community activity, enterprise adoption, and developer sentiment signals drawn from the developer-ecosystem-analytics dataset corpus.

1.4.2 Specific Objectives

1. To perform in-depth exploratory data analysis on the existing six-dataset corpus to identify statistically significant patterns and correlations relevant to technology trajectory.
2. To engineer predictive features from Stack Overflow activity trends, Fortune 500 company adoption depth, and developer sentiment survey scores.
3. To train and evaluate machine learning classification models (Logistic Regression, K-Nearest Neighbours, Random Forest, and XGBoost) for technology trajectory prediction.
4. To identify the strongest predictors of technology decline using SHAP feature-importance analysis.
5. To design and deploy an interactive Streamlit dashboard that visualises technology predictions, risk scores, and feature-importance explanations in real time.
6. To document the full project process, findings, and recommendations in a formal dissertation.

1.5 Significance of the Project

1.5.1 For Developers

Individual developers will gain access to a data-driven, evidence-based tool for evaluating whether a technology they are considering learning or deepening their expertise in shows signs of growth, stability, or impending decline. This directly addresses the risk of career setbacks caused by over-investment in technologies that quietly lose community and enterprise support.

1.5.2 For Organisations

Technology decision-makers, CTOs, engineering leads, and IT strategy teams, will benefit from an objective, explainable framework for assessing technology stack risk. TechPulse's risk scores and

feature-importance explanations will provide a structured basis for platform migration timing, technology refresh planning, and investment prioritisation.

1.5.3 For the Open-Source Community

Because TechPulse extends a publicly starred, open-source repository, the predictive layer and dashboard will be immediately available to the broader developer community upon release. This transforms a static descriptive analytics tool into a living, predictive resource for the global technology community.

1.5.4 For Academic Research

This project contributes a novel combination of multi-signal feature engineering, explainable machine learning, and enterprise-scale adoption data to the technology lifecycle prediction literature, a gap not yet addressed by existing studies reviewed in Chapter Two.

1.6 Scope

1.6.1 Functional Scope

TechPulse will: ingest and analyse the six existing datasets integrated into the developer-ecosystem-analytics repository; engineer predictive features from community, adoption, and sentiment signals; train and evaluate four classification models; surface predictions, risk scores, and feature-importance explanations (via SHAP) through a public Streamlit dashboard; and document the full process and findings in a dissertation.

1.6.2 Technical Scope

The system will use Python 3.10+, PostgreSQL 14+ (existing database), scikit-learn and XGBoost for modelling, SHAP for explainability, and Streamlit for the interactive dashboard. All development will take place on the existing developer-ecosystem-analytics repository infrastructure.

1.6.3 Geographical Scope

The dataset corpus covers global technology adoption patterns, drawing on Stack Overflow activity (worldwide), Fortune 500 enterprise stacks (USA-headquartered companies), and developer sentiment surveys (international respondents). Findings are therefore broadly applicable across geographies, with particular relevance for technology decision-makers in emerging markets, including East Africa, where access to evidence-based adoption guidance is limited.

1.6.4 Exclusions

The system will not: collect new primary data beyond the existing six datasets; provide financial or investment advice; guarantee prediction accuracy for technologies with insufficient historical signal data; or monitor technologies in real time beyond scheduled batch updates.

CHAPTER TWO: LITERATURE REVIEW

2.1 Developer Ecosystem and Technology Adoption Analytics

Research into open-source and developer-community platforms has increasingly used community activity as a proxy for technology health. Studies of GitHub starring behaviour have demonstrated that social signals, including stars, forks, and contributor counts, reflect genuine interest and adoption patterns rather than arbitrary popularity, supporting the use of community metrics as predictive features (Borges & Valente, 2018). Work analysing the evolution of social coding platforms has similarly demonstrated that contribution and activity patterns shift in measurable, trackable ways as projects and technologies mature, decline, or are superseded, directly motivating the trend-based features (such as Community Decay Rate and Growth Momentum Index) proposed in this project (Mens & Goeminne, 2012).

Large-scale empirical studies linking programming language choice to code quality outcomes provide further evidence that the technology a team adopts has measurable downstream consequences, reinforcing the practical importance of being able to forecast technology trajectory rather than only describe its current state (Ray et al., 2014).

2.2 Existing Approaches and Data Sources

2.2.1 Stack Overflow Developer Survey and Community Analytics

The annual Stack Overflow Developer Survey is one of the most widely used public datasets for understanding developer sentiment, technology satisfaction, and adoption intent at scale (Stack Overflow, 2023/2024). It is descriptive by design: it reports current sentiment and usage but does not itself produce forward-looking predictions. This project treats survey-derived sentiment as one input signal among several, rather than as a standalone forecasting tool. The limitation of relying solely on survey data is that it captures developer intention and opinion, not actual enterprise deployment behaviour, a gap this project addresses by combining survey sentiment with Fortune 500 adoption data.

2.2.2 Empirical Studies of Open-Source Project Evolution

Empirical case studies of open-source project activity have shown that prior activity patterns, including commit frequency, issue resolution rates, and community responsiveness, carry predictive signal about a project's future trajectory (Syer et al., 2013). These studies, however, generally focus on individual projects rather than technologies as a whole, and do not combine community signals with enterprise adoption data. This single-project limitation is a gap that TechPulse directly addresses by aggregating signals across all technologies in the corpus rather than analysing individual repositories in isolation.

2.3 Gap Analysis and Justification for TechPulse

The literature reviewed above confirms that: (a) community activity signals are valid proxies for technology health; (b) developer sentiment surveys provide useful but inherently descriptive adoption intent data; and (c) empirical project-level studies demonstrate that activity patterns carry predictive signal. However, no existing work reviewed combines all three signal families, community activity, enterprise adoption depth, and developer sentiment, into a single supervised classification framework covering technology categories (languages, frameworks, databases, and tools) rather than individual repositories.

Unlike the Stack Overflow survey or single-project evolution studies described above, TechPulse combines these three distinct signal families into a single supervised classification framework. It also moves beyond description to prediction, explicitly labelling technologies as growing, stable, or declining and explaining those predictions through SHAP-based feature importance. This

combination of multi-signal feature engineering, enterprise-scale adoption data, and explainable predictive modelling represents, to the author's knowledge, a novel contribution not addressed by the existing literature or tools reviewed above.

CHAPTER THREE: METHODOLOGY

3.1 Requirements Collection

All required data is already integrated into the existing developer-ecosystem-analytics repository, so no new primary data collection is needed. Functional and analytical requirements for the predictive layer will be gathered through:

- Review of the existing repository's twelve SQL analytics queries, to identify which derived metrics (e.g., Technology Health Score, Growth Momentum Index) can be reused or extended as machine learning model features.
- Consultation with the project supervisor, Dr. Rufus Gireka, to validate the target variable definition (growing / stable / declining) and model evaluation criteria.
- Examination of the six integrated datasets to assess data completeness, class distribution across the target variable, and the feasibility of the intended feature engineering steps.

These approaches were selected because the data infrastructure already exists; the research priority is correctly translating existing descriptive metrics into a predictive feature set rather than gathering requirements from new end users.

3.2 Development Methodology

This project will follow an iterative, experiment-driven development methodology suited to machine learning research, rather than a traditional software-only lifecycle methodology such as Waterfall. The methodology is structured around four iterative phases:

7. Exploratory Data Analysis (EDA), statistical profiling of all six datasets, identification of missing values and class imbalances, and correlation analysis to determine candidate features.
8. Feature Engineering, derivation of predictive variables from existing SQL queries and raw dataset fields, including Community Decay Rate, Growth Momentum Index, enterprise adoption breadth, and sentiment trajectory scores.
9. Model Training and Evaluation, iterative training of four classifiers (Logistic Regression, KNN, Random Forest, XGBoost), with 5-fold cross-validation, per-class confusion matrix analysis, and SHAP feature-importance ranking.
10. Dashboard Development and Deployment, construction of an interactive Streamlit dashboard presenting technology risk scores, trajectory predictions, and explainability outputs, deployed publicly via Streamlit Cloud.

Each phase is revisited iteratively as results from SHAP analysis and cross-validation reveal which features and models perform best, similar in spirit to Agile's short, evaluative cycles but adapted for a data science workflow.

3.3 Data Sources

Dataset	Source	Key Features Used
Stack Overflow Questions	Existing DB	Volume, scores, answers, closure rate by tag
Developer Sentiment Survey	Existing DB	Satisfaction, adoption rate, learning curve
Fortune 500 Tech Stacks	Existing DB	Company adoption depth per technology
Company Profiles	Existing DB	Parent company relationships, size, sector
Technology Metadata	Existing DB	Category, release year, lifecycle flags

Question–Company Mapping	Existing DB	Technology mentions by company over time
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3.4 Models to be Trained and Evaluated

Model	Justification	Evaluation Metrics
Logistic Regression	Interpretable baseline with coefficients that can be compared to SHAP values	Accuracy, F1-score (macro)
K-Nearest Neighbours	Non-parametric baseline; useful for detecting local clustering in feature space	Accuracy
Random Forest	Handles mixed feature types; provides native feature importance scores	Accuracy, F1, ROC-AUC
XGBoost	State-of-the-art performance on tabular datasets; gradient boosting with regularisation	Accuracy, F1, ROC-AUC

3.5 Evaluation Strategy

Model evaluation will use an 80/20 train-test split with stratified sampling to ensure balanced class representation across all three trajectory labels (growing, stable, declining). Five-fold cross-validation will be applied to assess model robustness and reduce variance in performance estimates. Evaluation will include per-class precision, recall, F1-score, macro-averaged F1, ROC-AUC scores, and confusion matrix analysis. SHAP values will be computed for the best-performing model to produce interpretable, ranked feature-importance explanations.

APPENDIX A: RESOURCES

SN	Resource	Specification	Usage
1	Database	PostgreSQL 14+	Existing data storage and SQL analytics
2	Programming Language	Python 3.10+	Data manipulation, feature engineering, modelling
3	ML Libraries	scikit-learn, XGBoost	Model training and evaluation
4	Explainability Library	SHAP	Feature importance and model interpretation
5	Visualisation Libraries	Matplotlib, Seaborn, Plotly	EDA charts and result plots
6	Dashboard Framework	Streamlit	Interactive prediction interface
7	Computer	8 GB+ RAM, mid-range CPU	Development, model training, and storage
8	Version Control	Git / GitHub	Code management (existing repository)
9	Documentation Tool	Jupyter Notebooks	Reproducible analysis and reporting

APPENDIX B: BUDGET

SN	Resource	Quantity	Availability	Estimated (KShs)	Actual (KShs)
1	Internet connection	40 hours	Own	3,000	
2	Cloud/dashboard hosting (Streamlit Cloud)	2 semesters	Free tier	0	
3	Flash disk / external backup	1	Own	1,500	
4	Computer	1	Own	0 (already owned)	
5	Printing & binding (dissertation)	2 copies	To purchase	2,500	
6	Contingency	—	—	1,000	

APPENDIX C: PROJECT SCHEDULE

Semester 1 – Data, EDA and Modelling

Week(s)	Activity	Deliverable
1–2	Literature review: technology lifecycle prediction, ML on developer data	Annotated bibliography
3–4	Deep EDA on all six datasets; statistical correlation analysis	EDA report and visualisations

5–6	Feature engineering: derive all ML features from existing SQL queries	Feature matrix (CSV)
7–9	Train baseline models: Logistic Regression and KNN	Baseline model results
10–12	Train advanced models: Random Forest and XGBoost; hyperparameter tuning	Final trained models
13–14	Model evaluation, SHAP analysis, feature importance ranking	Evaluation report
15–16	Semester 1 progress report and supervisor review	Progress report

Semester 2 – Dashboard, Dissertation and Presentation

Week(s)	Activity	Deliverable
1–3	Build Streamlit dashboard: tech risk scores, predictions, trend charts	Live dashboard
4–5	Deploy dashboard (Streamlit Cloud)	Public URL
6–8	Write dissertation chapters 1–3: Introduction, Literature, Methodology	Draft chapters
9–11	Write dissertation chapters 4–5: Results, Discussion and Recommendations	Draft chapters
12–13	Full dissertation review, corrections, formatting	Complete draft
14	Final submission	Bound dissertation
15–16	Presentation preparation and viva	Presentation slides

APPENDIX D: PROJECT RISKS AND MITIGATIONS

SN	Risk	Effect	Mitigation
1	Class imbalance in growing/stable/declining labels	Biased model performance; misleading accuracy	Use stratified sampling; report F1-score and ROC-AUC alongside accuracy
2	Insufficient historical signal for newer technologies	Unreliable predictions for some categories	Flag low-confidence predictions; exclude technologies below a minimum data threshold
3	Hardware failure or theft	Delay; lost code/data	Regular commits to GitHub; periodic backups to flash disk/cloud storage
4	Dashboard hosting downtime or free-tier limits	Dashboard unavailable for evaluation	Maintain a local fallback build; document alternative hosting option
5	Scope creep beyond defined project boundaries	Missed deadlines	Refer back to Section 1.6 Scope; defer extras to future work

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